CHAPTER 6

Blockchain for IoT-based medical delivery drones: state of the art, issues, and future prospects

Partha Pratim Ray and Dinesh Dash
Department of Computer Science and Engineering, National Institute of Technology Patna, Patna, Bihar, India

Nomenclature

AIRA	Autonomous intelligent robot agent
ATC	Air traffic control
BCoT	Blockchain of things
BVLOS	Beyond visual line of sight
CC	Command and control
eVTOL	Electric vertical takeoff and landing
HIRO	Healthcare integrated rescue operations
IDS	Intrusion detection system
IoDT	Internet of drone things
IPFS	Interplanetary file system
KPI	Key performance indicator
LoS	Line of sight
MAS	Multiagent systems
SDU	University of Southern Denmark
SOP	Standard operating procedure
UAS	Unmanned aerial system
UHF	Ultra high frequency
UPDWG	UAV for payload delivery working group
VHF	Very high frequency
VTOL	Vertical takeoff and landing

1. Introduction

The internet of things (IoT) refers to the ecosystem that allows billions of distributed things to exchange information by using sensors, actuators, microcontrollers, and network backhaul (Pawar et al., 2021; Hemalatha, 2021; Frikha et al., 2021; Sadeeq and Zeebaree, 2021). Since its inception, IoT

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has been successfully deployed in a wide range of smart applications, e.g., healthcare, agriculture, transportation, military, and industry (Ratta et al., 2021; Gul et al., 2021; Liang and Ji, 2021). It has evolved due to convergence of a multitude of heterogeneous technologies comprising pervasive computing, ubiquitous computing, embedded systems, and wireless sensor networks (Chinaei et al., 2021; Alzubi, 2021; Mourya et al., 2021; Guo et al., 2021). The main aim of IoT is to automate things with an aware workflow. For example, a pulse sensor can measure the pulse rate of a person and send the information to a remote cloud for further processing. Robotic elements are integrated with the IoT-based infrastructure to deploy in the industry 4.0 use cases (Gimenez et al., 2021; Lv and Singh, 2021; Philip et al., 2021). Recently, drones have been getting significant attention from many organizations and startups to use them as a key component in some prospective business models (Saranya, 2021; Maitra et al., 2021; Abbas et al., 2021; Majeed et al., 2021; Bhuvaneshwait et al., 2021), especially for the delivery of medical items to remote outdoor locations. A drone is designed to serve remote locations without a human pilot. Drones are equipped with communication technologies to establish connections with the ground command and control (CC stations) (Kumar et al., 2021; Kumar and Tripathi, 2021; Shukla et al., 2021; Chen et al., 2021; Jeong et al., 2021; Velmurugadass et al., 2021; Chittoor et al., 2021). Ordinarily a drone is an autonomous thing that was earlier used for military applications. Nowadays, drones are easily available in various forms, factors, and cost. Drones can be found in following types: multirotor, fixed wing, single rotor, and hybrid vertical takeoff and landing (VTOL) (Nehme et al., 2021; Miglani and Kumar, 2021; Singh et al., 2021; Shynu et al., 2021; Evita et al., 2021). Drones can be accommodated with high resolution cameras and sensory elements to capture images, collect information, monitor target zones, and provide control signals. Drones can fly at different heights above sea level (600, 1500, 3000, and 5000 m) based on design and regulatory constraints (Jamil et al., 2021; Kiran et al., 2021; Saxena et al., 2021; Aruna, 2021; Svedin et al., 2021). Propulsion can be controlled by internal combustion engines, rotary engines, and lithium-polymer battery assembly (Ribeiro et al., 2021). They can be designed as real-time things that are accompanied with smart, embedded microcontroller boards, e.g., Raspberry Pi and Beagle boards (Rominger et al., 2021; Homier et al., 2021).

Smart medical item delivery is an important use case of the smart healthcare domain that can be upgraded with the inherent capability of autonomy and real timeliness (Liu and Szirányi, 2021). Drones are being considered the key player of this application to deliver medical objects to remote locations (Martins et al., 2021). Drones can be accommodated with blood, pathology samples, vaccines, smart IoT-based health sensors, and related medical items to support patients located distant from a standard medical facility such as a hospital (Da Xu et al., 2021; Yaqoob et al., 2021; Comtet and Johannessen, 2021). Such an approach can benefit the health condition of citizens all over the world, especially in low-income or developing countries (Tarr et al., 2021). However, there exists some key research issues that can hinder the process of adoption of drones for the purpose of medical delivery (Saraf et al., 2021). For example, existing medical delivery faces the following challenges: user data privacy, data security, tampering of sensitive medical records, delay in package delivery, and lack of transparency.

Blockchain technology can be used to minimize such issues along with IoT and drones for medical delivery (Shuaib et al., 2021). A blockchain is a growing list of digital records, i.e., blocks. Each block of a blockchain contains a timestamp, transaction history, nonce, and hash values to point to the previous block (Iqbal et al., 2021). Thus, a change in a random block would change the hash associated with it, leading to mismatch with other blocks of the chain (Chelladurai and Pandian, 2021). Blockchain uses peer-to-peer network architecture to share data in distributed ledgers. A node that is acting as a ledger needs to adhere to the protocols to validate the blocks before using it (Gul et al., 2021). The blockchain allows the decentralization, transparency, and consensus algorithm mechanisms to cater tamper-proof, immutable data storage and access (Fatokun et al., 2021). Blockchain technology is used in many applications such as financial services, cryptocurrencies, supply chain management, anticounterfeiting, video games, energy trading, and healthcare (Pavithran et al., 2021; Ali et al., 2021; Santos et al., 2021; Farahani et al., 2021; Awan et al., 2021). Blockchains are normally classified into three types: public, private, and hybrid. Public blockchain has fewer restrictions over accessing the blocks of the chain by any user. Private blockchains are permissioned and require an entity to authenticate before accessing the blocks (Attraian and Hashemi, 2021; Yang et al., 2021; Khubrani, 2021). Distributed ledger technology refers to the use of private blockchains. Hybrid blockchains have a combination of centralized and decentralized attributes. There exists another special type of blockchain, called sidechain, that can run in parallel to the primary blockchain. Blockchains should ensure the use of consensus algorithms that enable distributed coordination between decentralized entities (Serrano, 2021; Soni and Singh, 2021; Alqaralleh et al., 2021;

Pathak et al., 2021; Panda et al., 2021; Meier et al., 2021). Sometimes, voting is done among the participating entities to reach agreement. Fig. 6.1 presents drone-based medical delivery using IoT and blockchain.

The major research questions of this study are as follows:
- RQ1: Can drones serve the medical industry?
- RQ2: Are drones compatible with health services?
- RQ3: What type of drones are available for medical aid?
- RQ4: What is the comparative structure of such drones?
- RQ5: What are important use cases for medical drone delivery?
- RQ6: What are the key challenges of medical delivery drones?
- RQ7: Can we provide future direction in this context?

Drones can be enabled with blockchains and IoT to enrich the prospects of medical delivery opportunities to remote places. In this paper, we discuss the prospects of using drones as a key component in the purpose of medical item delivery. First, we discuss the existing drone suppliers for medical delivery and compare such drones based on model name, method of flight, propulsion, power source, payload capacity, payload volume, maximum flight time with full payload, radio links, delivery method, and role. Second, we present the status of drone-based medical delivery services based on an in-depth review of available literature. Third, we depict major challenges associated with the present scenario and leverage future directions.

The key contributions of the chapter are as follows:
- to evaluate available drones for medical delivery
- to assess the state of the art on drone-based blockchain-IoT enabled medical delivery
- to present key research challenges and provide future prospects

The rest of the chapter is organized as follows. Section 2 presents related works on the given topic of discussion. Section 3 evaluates drones under deployments for medical delivery. Section 4 provides an in-depth review on classifications of medical delivery approaches based on blockchain and IoT. Section 5 discusses research challenges and delivers future directions. Section 6 concludes the chapter.

2. Related works

Blockchain can be used with 5G technology for demonstration of various applications. Nguyen (Nguyen et al., 2020) described that cloud and edge computing are integrated with network function virtualization for enabling 5G-aware service provisioning. Device-to-device communication is used to

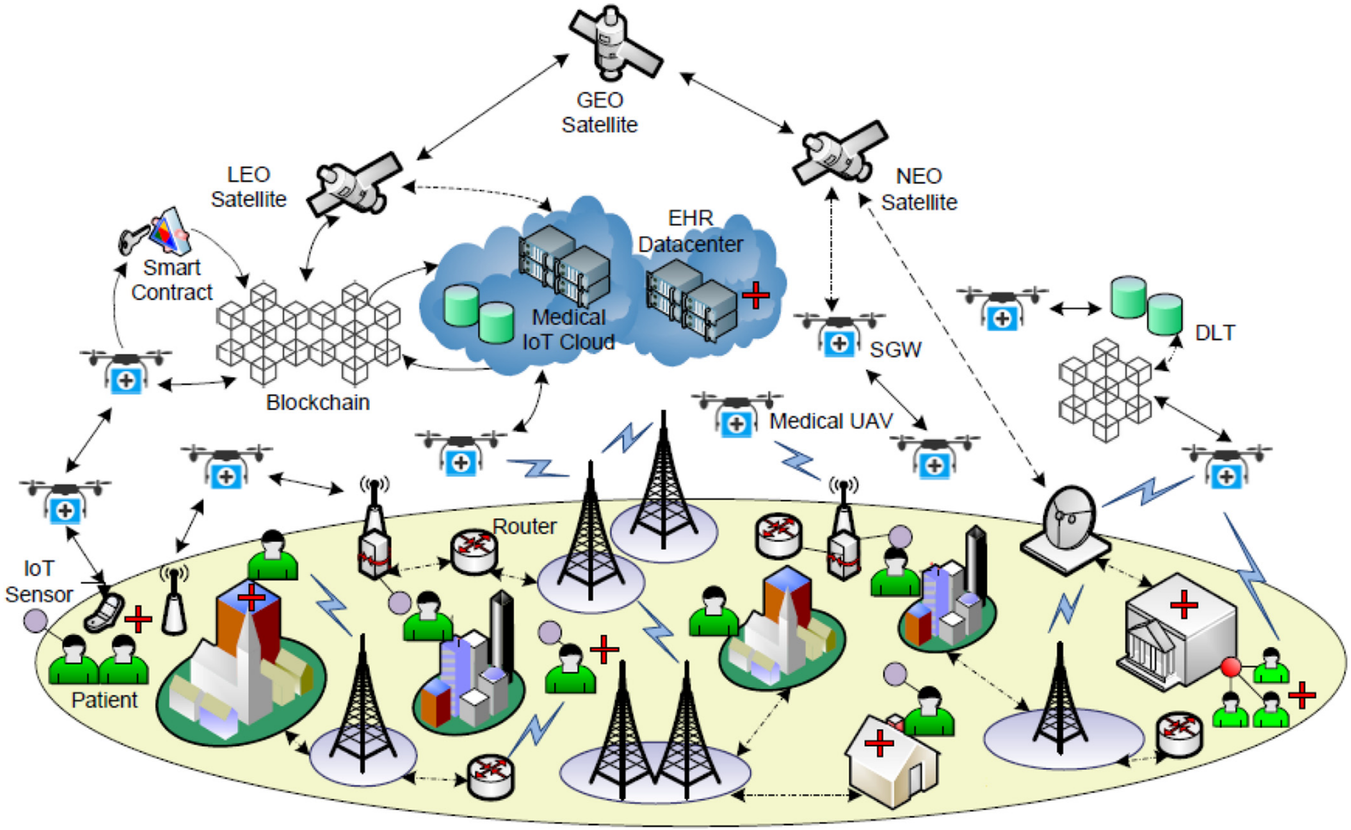

Figure 6.1 Drone-based medical delivery architecture using IoT and blockchain.

showcase the network slicing scheme for efficient management of numerous use cases using 5G-based blockchain. This study discusses vital notions of 5G-assisted blockchain deployment scenarios that includes resource management, federated learning approach, privacy concerns, data sharing, spectrum management, and network virtualization. This work presents the way the UAVs should be aligned with the 5G-blockchain ecosystem. We find that this article is silent about the medical delivery-specific domain of application.

Beyond 5G (B5G) technology can be integrated with a blockchain and drones for alleviating the green internet of things (Alsamhi et al., 2021a,b). The Alsamhi study shows that drones can fly near IoT-based devices. Thus, the drones can be useful to minimize the overall power consumption by the IoT devices. This study showcases various approaches where IoT, blockchain, and drones can fit together for supporting a wide range of applications, such as, monitoring, surveillance, transportation, public safety, healthcare, and environmental prevention. However, this work does not deal with the medical delivery aspect, where drones can be highly involved.

We find that IoT can improve industrial applications, especially industry 4.0, by using smart sensors, actuators, microcontrollers, and decision-making algorithms. An in-depth survey on IoT and its implications on blockchain is presented in a work by Dai et al. (2019). This work synthesizes IoT as a blockchain of things (BCoT) to amalgamate with futuristic use cases. We notice this study did not specify drone-based medical delivery approaches. A related tertiary study on blockchain-IoT was discussed in Xu (Xu et al., 2020). However, no mention of drone-based medical delivery was found in that study either.

We observe that McGovern (2020) discussed several factors and types of data needed by the blockchain-UAV ecosystem. It presented how drones, operators, and hospitals together can support healthcare applications. This study proposed a case study where organ supplying was envisaged with help of blockchain-assisted drones. In this case study, hospitals first write organ-supplying information in the blockchain, which is then read by the air traffic control (ATC) to allow drone operators to start the organ delivery. A drone then ships the organ to the destination hospital where authorized staff receive the organ after validating the transaction process over the blockchain.

In Ferrag and Maglaras (2019), the DeliveryCoin, a cryptocurrency, was proposed to allow blockchain-based drone-assisted delivery of goods. DeliveryCoin uses a strong Diffie-Hellman algorithm to achieve a secure consensus algorithm among various drones. An intrusion detection system (IDS) was integrated with the proposed blockchain ecosystem to allow 5G technology for establishing communication between drones and control centers. A machine learning–based IDS was proposed by Khan et al. (2021) where predictive analytics was incorporated to detect IDS in a decentralized fashion.

Deep learning (DL) technique is a popular way to make complex decisions based on given input. In Gumaei et al. (2021), such an architecture was proposed that can deal with DL-enabled decision-making capability in drones. This study presents that 5G can be considered an important communication technology to perform delay-aware drone identification and flight mode detection on the go. We could not find the medical delivery aspect in that study. In another work (Fraga-Lamas et al., 2019), a review on IoT-based DL techniques was conducted to make the UAVs self-aware of obstacles and avoid collisions on the fly. This work presented an architecture that considered DL-based unmanned aerial system (UAS) for futuristic application deployments.

Trust management was included in a novel drone-based blockchain system for measurement and protection of critical infrastructure (Barka et al., 2019). Besides performing a survey, the major contribution of this work was to propose a blockchain-based framework to confirm the trust and security of IoT-integrated UAVs.

A feasibility analysis of financial risk was performed (Hiebert et al., 2020) while considering drones as key enablers of smart healthcare applications. However, there was no mention about the possibility of medical delivery by drones under a blockchain-IoT environment.

Privacy preservation was mandated by Wu et al. (2021), where a blockchain-based 5G-enabled infrastructure was proposed. The study was designed around the pandemic situation where collection and dissemination of large data can be seamlessly done by drones. This study presented an approach to secure the information and make the drone communication aware of privacy.

We observe that Islam et al. (2021) presented a blockchain-aware system that can supervise some decision-making tasks with help of artificial intelligence technique. This study was conducted while considering a pandemic situation like COVID-19. This study discussed the internet of drone things (IoDT) to revolutionize a smart healthcare application during times of epidemic. It employed a lightweight blockchain platform on the drones to collect IoT-based data from remote locations and monitoring of

COVID-19 standard operating procedure (SOP) with poor network connectivity. A two-phase security algorithm was deployed to adopt the consensus algorithms under a resource-constrained scenario.

In another study (Fernández-Caramés et al., 2019), we find an approach to deal with big data–driven supply chain management by using UAVs and blockchain. The main aim of the study was to design an industry 4.0 warehouse that can be monitored for current status about the inventory. Further, the component on traceability was implied to make the whole supply chain feasible, cost effective, and reliable.

A discussion on UAV and blockchain was made by Kosuda et al. (2020). The study presented the utilization of commercial UAS operators under the aegis of a blockchain-based ecosystem. It was envisaged that the UAS can be aligned with unmanned traffic management for incorporation of IoT, 5G, and artificial intelligence toward smart, UAV-assisted operation management. A similar discussion was found in Han et al. (2021) where emergence of drone-aware trends was investigated. More emphasis was given to the utilization of mobile edge computing on top of drones on the fly so blockchain technology can be used to bring a novel application paradigm.

In Ray and Nguyen (2020), an in-depth review was conducted to investigate the status of blockchain-assisted medical drone delivery under the 5G-IoT ecosystem. This study presented a multimodal approach where medical drone delivery could be possible by using 5G and IoT. A satellite-based 5G-IoT drone architecture was proposed that was supported by the blockchain to maintain the logs of medical delivery.

2.1 Research gap

We found several research gaps after extensive surveys of related works. First, we find that the majority of the literature is silent about the drone suppliers for medical delivery operations. Second, existing literature does not conform to the use of blockchain for medical delivery via drones. Third, tere are minimal investigations on IoT-edge-centric ecosystem design for medical drone delivery. Finally, there is a lacuna in prospective use cases in this context.

2.2 Lessons learned

We learned that related works covered blockchain, 5G, IoT, and drone-aware studies. It was noticed that most of the works except Ray and Nguyen (2020) did no investigation of drone-based medical delivery.

The novelty of this paper can be expressed as follows: explicit investigation of drone suppliers for medical delivery, review of present status on drone-based medical delivery, implications of blockchain and IoT on the present situation of drone-aware medical delivery, and assessment of existing challenges.

3. Drone suppliers for medical delivery

Several drone suppliers are actively working toward delivery of medical items and services to different parts of the globe. Such drone suppliers have come up with a new consortium called the UAV for Payload Delivery Working Group (UPDWG). The UPDWG includes multiple stakeholders belonging to academic institutions, nonprofit organizations, lenders, private companies, and UAV suppliers. UPDWG allows four-phase inclusion of drones consisting of technology demonstration, safety and feasibility testing, effectiveness measurement, and impact expansion. Most of the medical delivery services are confined within a few countries such as Ghana, Madagascar, Malawi, Rwanda, Vanuatu, and Democratic Republic of the Congo. However, some drone suppliers are in the initial phases of UPDWG and are testing their prototypes in following countries: South Africa, Sweden, Italy, United Kingdom, Mozambique, Canada, Nepal, Botswana, France, Kazakhstan, Germany, Netherlands, Senegal, Tanzania, and United States of America (UPDWG, 2021). Such drones can carry different types of payloads that include saline bags, blood, medications, vaccines, first aid kits, diagnostic samples, tuberculosis sputum samples, syringes, defibrillators, oxytocin, and essential portable medical products. We present a list of the drone suppliers that have completed all four phases of testing and are now included in the official technology database of the UPDWG (MD3, 2021).

- Matternet

  Matternet is a Berlin-based drone supplier that deploys beyond visual line of sight (BVLOS). It provides an end-to-end (E2E) Matternet M2 drone for facilitation of medical item delivery between health stations. It uses Matternet station and Matternet cloud platform to perform tasks such as, monitoring, CC, customer response, and on-demand routing (Matternet, 2021).
- Vertical Technologies

  Vertical Technologies uses VTOL drones, which are available in various models to perform mapping, viewing, inspection, and cargo operations. It depends on the DeltaQuad VTOL drones that can fly up to 150 km by using an auxiliary battery with 1.2-kg payload. DeltaQuad drones can fly both in snowy and rainy conditions and have less than 1-min field assembly duration (Vertical, 2021).

- DJI

  DJI uses DJL Air 2S aircraft with 595-g payload weight. It has a maximum ascent speed of 6 m/s both in S and N modes. It can fly above 5000 m sea level with hovering time of 30 min. In a no wind condition, the drone can travel 18.5 km. The drone has a high accuracy hovering range with foldable aircraft arms. It uses one CMOS camera sensor with 20 megapixels and 100—3200 ISO automatic video range. It provides smartphone apps for CC of the drone with advanced safety and obstacle sensing (DJI, 2021).

- Cloudline

  Cloudline drone suppliers can serve with 10-kg medical payload and around 50-km flying range. Such drones can attain 9 knots during headwinds and 40 km/h during cruise speed. It uses a Cloudship I drone with an electric rotor with a built-in ultra high frequency (UHF) radio communication module. It uses landing-based medical package delivery (Cloudline, 2021).

- Germandrones

  Germandrones uses a reliable flight management system having autopilot, ground station, and data interconnection systems. It provides live video cameras including Nextvision Colibri II, Merio Temis XL, and photo cameras (Sony Alpha series, MicaSense Altum, PhaseOne iXM-100). Several fixed wing drones are used to enable long flight time and distances (Germandrones, 2021). It uses 4G data transmission technologies for high-quality video data propagation. It deploys a Songbird drone that can provide VTOL for medical delivery.

- Phoenix-Wings

  Phoenix-Wings supplies the PWOne, which is the smallest VTOL drone that can carry 500 g of payload for 40 km. Manta Ray is another drone belonging to the Phoenix-Wings that is used for long-range (55—95 km) heavy-duty (10-kg) package delivery (Phoenix, 2021).

- UAV Factory

  UAV Factory deploys the Penguin B, a fixed wing drone with an internal combustion engine. It has satellite communication modules for establishing data transmission with the drone stations. It uses parachutes to deliver medical items to the destination. Penguin C is a type of long-endurance (16–20 h) drone that can fly with a line-of-sight (LoS) range of 100 km. It is capable of carrying 2 kg of payload with a maximum cruise speed of 19–22 m/s at 5000 m above sea level (UAV, 2021).

- Vayu

  Vayu provides the VTOL-based drones for moderate payload delivery. There exist two models (X5 and G1) that can work on battery or petrol power sources (Vayu, 2021). X5 uses UHF radio and G1 uses satellite communication, 3G, and 4G LTE. Average endurance of travel is 480 h with a full payload having 2–5 kg.
- NextWing

  NextWing supplies two types of drones (NXT 1 and NXT 11) for delivery, survey, and security applications. They contain insulated cargo that can hold medical shipments and safeguard it from external environmental impacts (Nextwing, 2021). They comprise high-resolution cameras (e.g., 42 megapixel, SLR sensors) to capture images and sense obstacles. NXT 1 can carry 1 kg of payload, whereas NXT 11 can deliver 11 kg of payload to the destination. The speed is around 90–95 km/h with a long range (70–190 km).
- Swoop Aero

  Swoop Aero supplies scalable drones to serve medical delivery anywhere on the planet. Swoop Aero supports the delivery of pathologies, pharmaceuticals, vaccines, and urgent blood to locations where they are needed (Swoop, 2021). It depends on the digital twin of the deployed drone. The underlying system helps to communicate the physical hardware to the digital twin.
- Wingcopter

  Wingcopter provides a multitude of applications based on the two drones (W198 and W178). It can support mapping, inspection, intersite logistics, and medical item delivery. W178 and W198 have a wingspan of 178 and 198 cm, respectively. W178 can hover 100 km with 2 kg of payload and 120 km without any payload (Wingcopter, 2021). W198 can fly 110 km without payload and 75 km with a 5-kg payload. Both W178 and W198 can fly 5000 m above sea level. Their average fixed wing speed is 148 km/h. Both support cellular and satellite telemetry for ground station connectivity.
- Zipline

  Zipline is one of the most popular medical delivery drone suppliers in the world. It has different generations of drones (first, second, and third) that are based on fixed wing design and electric motors (Zipline, 2021). It uses parachutes to deliver medical items to the destination. Zipline is used in both deliveries and integrated on-demand delivery services. It has served many flights in Ghana in the past. Zipline is well known for its high resilience, scale, speed, and range.

- University of Southern Denmark

  University of Southern Denmark (SDU) has designed both fixed and multirotor health drones for medical delivery. Recently, it originated the GENIUS project to deploy 5G-based drone communication networking by following the BVLOS mechanism. Currently, the SDU drones can carry lightweight payload (100–300 g) with battery-supported endurance (SDU, 2021). It can fly 20–45 km with payload to deliver the package to the destination. It is supported with an electric motor and can communicate via UHF radio.
- Aurora Aerial

  Aurora Aerial deploys different versions of drones (AAC-1000, AAT-1000, AAT-1200, AAF-1000) for medical delivery. The drones can carry 5–36 kg of payload with endurance of 40–130 km (Aurora, 2021). The majority of their drones follow the multirotor design, whereas AAF-1000 uses helicopter design. Both AAC and AAT models fly on the basis of electric motors, whereas AAF can fly on internal combustion engines. All the drones use landing facilities for medical delivery.
- Healthcare Integrated Rescue Operations (HIRO)

  HIRO is a telemedicine supplier that depends on its drone the HIRO for medical delivery. The HIRO drones are internally guided and use the BLOV mechanism. Such drones are useful for medical search and rescue operations. It uses a multirotor design for a more than 40-mile flight distance (HIRO, 2021).
- AerialMetric

  Aerial Metric deploys the Savior 330 drone, which is an electric VTOL (eVTOL) fixed wing UAV that runs on an electric motor. It can carry 10 kg of payload with 130 km of flight distance. It also uses parachutes to deliver the medical items to the destination. The drone can fly at 5000 m above sea level and the speed is 150 km/h (Aerialmetric, 2021). When the payload weight is minimized, it can fly for 300 km with 3-h battery backup. Besides, it is equipped with UHF radio, 4G LTE, and satellite communication modules.

- Quantum Systems

  Quantum systems uses three major designs of drones (Trinity F90+ UAS, Vector UAS, and Scorpion UAS). All the drones are equipped with eVTOL techniques and electric motors (Quantum, 2021). Trinity F90+ comprises RGB and NDVI payload sensors for mapping of areas. It follows 2.4 GH telemetry and has a 7-km CC range. The Vector can fly for 120 min covering a 15-km range. The Scorpion is built on top of a tricopter UAV design and has 0–15 m/s cruise speed and 45 min of flight.
- Windracers

  Windracers deploys a fixed wing UAV called ULTRA (Windracers, 2021). ULTRA flies based on the internal combustion engine fueled by petrol. It can carry 100 kg of payload for a 240-km range. It comprises UHF radio, very high frequency (VHF) radio, satellite communications, and 4G LTE for establishing connectivity with base stations. It delivers medical packages by using both parachute and landing approaches. It can capture multispectral imagery besides collecting samples from the patients.
- RigiTech

  RigiTech drones (RiGiOne) are controlled by the Rigi cloud management system. Such drones are equipped with VTOL specification and can fly 75 km (Rigitech, 2021). The payload volume is 15 L, which is fueled by a battery. It can carry 2 kg of solid payload to a landing facility. It can perform both delivery and collection of samples.
- Avy

  Avy uses their flagship drone, Aera, that is a VTOL fixed wing system. It is fueled by a battery to run an electric motor. Maximum payload is 1.5 kg for 55-km range (Avy, 2021). It depends on a landing feature to deliver the shipments. Figs. 6.2 and 6.3 present various drones. Table 6.1 compares medical delivery drones.

4. Blockchain-IoT aware drone-based medical delivery

In this section, we list the most important notions of blockchain-assisted medical delivery with the help of IoT and drones. We present key classifications of drone-aware medical delivery such as last mile delivery, collaborative and multiagent delivery, epidemic support delivery, forensic delivery,

Figure 6.2 (A) DeltaQuad VTOL applications; (B) mapping; (C) viewing; (D) inspection; (E) cargo delivery; (F) Manta Ray; (G) Penguin C; (H) Vayu X5; (I) NextWing NXT 1; (J) Swoop Aero drone is fitted with medical shipment; (K) Swoop Aero digital twin management architecture. *(Courtesy of (A) Vertucal, 2021. https://www.deltaquad.com/. (Accessed 26 June 2021), (H) Vayu, 2021. https://vaayudrones.com/. (Accessed 26 June 2021), (I) Nextwing, 2021. https://flynextwing.com/. (Accessed 26 June 2021), and (K) Swoop, 2021. https://swoop.aero/. (Accessed 26 June 2021)).*

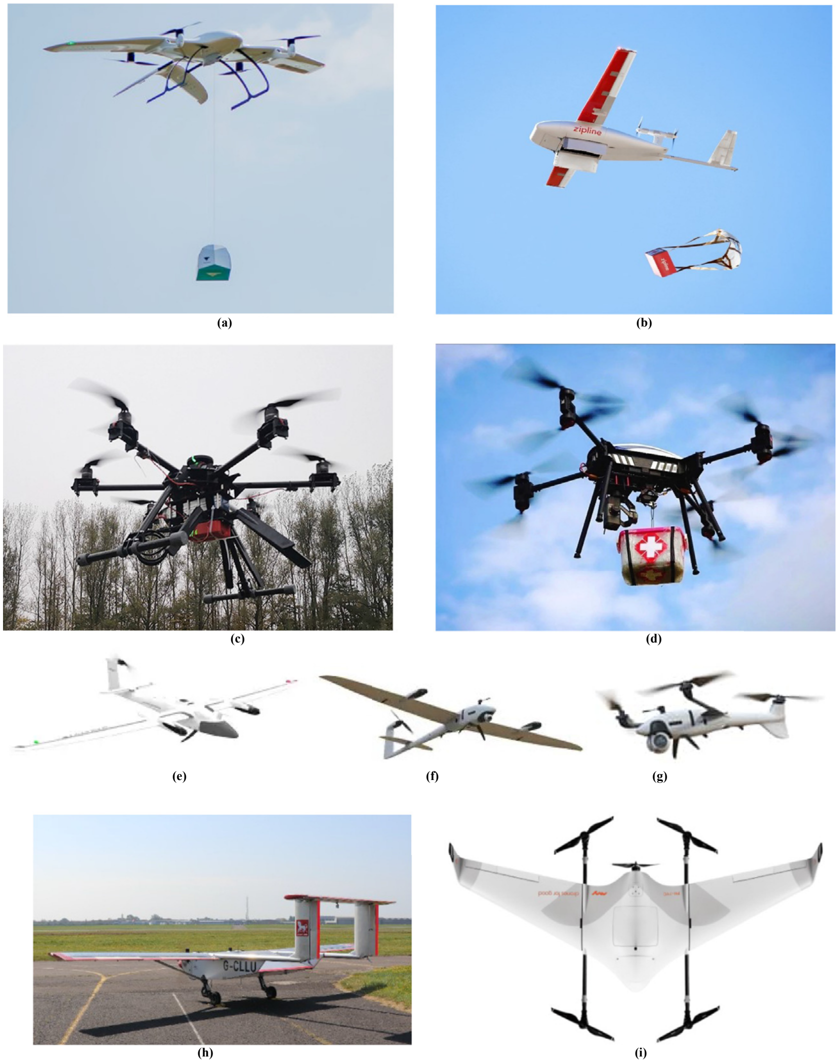

Figure 6.3 (A) Wingcopter; (B) Zipline; (C) SDU drone; (D) Aurora Aerial AAC-1000 drone; (E) Trinity F90+; (F) Vector; (G) Scorpion; (H) Windracer's ULTRA; (I) Aera drone. *(Courtesy of (A) Wingcopter, 2021. https://wingcopter.com/. (Accessed 26 June 2021), (B) Zipline, 2021. https://flyzipline.com/. (Accessed 26 June 2021), (C) SDU, 2021. https://www.sdu.dk/en/forskning/sduuascenter. (Accessed 26 June 2021), (D) Aurora, 2021. https://www.auroraaerial.aero/. (Accessed 26 June 2021), (E–G) Quantum, 2021. https://www.quantum-systems.com/. (Accessed 26 June 2021), (H) Wingcopter, 2021. https://wingcopter.com/. (Accessed 26 June 2021), and (I) Swoop, 2021. https://swoop.aero/. (Accessed 26 June 2021)).*

Table 6.1 Comparison of drones for medical delivery.

Drone supplier	Model name	Method of flight	Propulsion	Power source	Payload capacity (g)	Payload volume (cm)	Maximum flight time with full payload (min)	Radio links	Delivery method	Role
Matternet	M2	Multirotor	Electric motor	Battery	3	22 × 14 × 14	10	Autonomous	Landing	Delivery only
Vertical Technologies	Delta Quad	VTOL fixed wing	Electric motor	Battery	1	16 × 12 × 7	10	LTE (3G/4G)	Landing	Delivery only
DJI	M600 plus	Multirotor	Electric motor	Battery	3	22 × 14 × 14	12	LTE (3G/4G)	Landing	Delivery and collection
DJI	M600	Multirotor	Electric motor	Battery	6	22 × 14 × 16	29	Autonomous	Landing	Delivery, collection, video, still and multispectral imagery
Cloudline	Cloudship I	Other	Electric motor	Battery	4.5	470 × 180 × 180	208	UHF radio	Landing	Delivery and collection
Germandrones	Songbird	VTOL fixed wing	Electric motor	Battery	4.5	15 × 12 × 20	60	UHF radio	Landing	Delivery, collection, video, and multispectral imagery
Phoenix-Wings	Manta Ray SR	VTOL fixed wing	Electric motor	Battery	10	39.5 × 36.5 × 21	45	UHF radio	Landing	Delivery and collection
UAV Factory	Penguin B	Fixed wing	Internal combustion engine	Petrol	2	18 × 13 × 8	120	Satcom	Parachute	Delivery only
Vayu	X5	VTOL fixed wing	Electric motor	Battery	2	55 × 15 × 14		UHF radio	Landing	Delivery and collection

Vayu	G1	VTOL fixed wing	Hybrid	Battery and petrol	5	22.6 × 22.2 × 14	480	Satcom, LTE (3G/4G)	Landing	Delivery and collection
NextWing	VTL3D	VTOL fixed wing	Electric motor	Battery	1	10 × 10 × 12			Landing	Delivery and collection
Swoop Aero	Kookaburra	VTOL fixed wing	Electric motor	Battery	3	19 × 12 × 9	60	Satcom, LTE (3G/4G)	Landing	Delivery and collection
Wingcopter	178 Heavy Lift	VTOL fixed wing	Electric motor	Battery	6	178 × 132 × 52	22.5	LTE (3G/4G), UHF radio	Winch	Delivery only
Wingcopter	178 Heavy Lift Skyports	VTOL fixed wing	Electric motor	Battery	6	81 × 37 × 20	25	LTE (3G/4G), UHF radio, VHF radio, and Satcom	Landing	Delivery and collection
Zipline	Zip UAV 1st Generation, Zip UAV 2nd Generation, Zip UAV 3rd Generation	Fixed wing	Electric motor	Battery	1.75	25 × 25 × 15	180	LTE (3G/4G)	Parachute	Delivery only
University of Southern Denmark	HealthDrone Fixed Wing	Fixed wing	Electric motor	Battery	0.1	12 × 12 × 3	45	UHF radio	Landing	Delivery only
University of Southern Denmark	HealthDrone Multirotor	Multirotor	Electric motor	Battery	0.3	10 × 10 × 5	20	UHF radio	Landing	Delivery and collection
Aurora aerial	AAT-1200 Thunderbolt	Multirotor	Electric motor	Battery	9.1	40 × 40 × 40	20	LTE (3G/4G)	Landing	Delivery only
Aurora aerial	AAC-1000 Thunderbolt	Multirotor	Electric motor	Battery	5	45 × 40 × 40	40	LTE (3G/4G)	Landing	Delivery and collection

Continued

Table 6.1 Comparison of drones for medical delivery.—cont'd

Drone supplier	Model name	Method of flight	Propulsion	Power source	Payload capacity (g)	Payload volume (cm)	Maximum flight time with full payload (min)	Radio links	Delivery method	Role
Aurora aerial	AAF-1000 Mfit	Helicopter	Internal combustion engine	Diesel	36	N/A	120	Autonomous	Landing	Delivery, collection, video, still and multispectral imagery
AerialMetric	Savior 330	VTOL fixed wing	Electric motor	Battery	10	41 × 41 × 8.5	130	LTE (3G/4G), UHF radio, Satcom	Parachute	Delivery and video
Quantum systems	Trinity F90+	VTOL fixed wing	Electric motor	Battery	0.5	12.5 × 8.2 × 10.8	90	UHF radio	Landing	Delivery only
Quantum systems	Tron	VTOL fixed wing	Electric motor	Battery	2	24.9 × 8.7 × 9.7	60	UHF radio	Landing	Delivery only
Windracers	ULTRA	Fixed wing	Internal combustion engine	Petrol	100	N/A	240	LTE (3G/4G), UHF radio, VHF radio, Satcom, autonomous	Parachute and landing	Delivery, collection, video, still and multispectral imagery
RigiTech	RigiOne	VTOL fixed wing	Electric motor	Battery	2	45 × 21 × 16	30	LTE (3G/4G), UHF radio	Landing	Delivery and collection
Avy	Aera	VTOL fixed wing	Electric motor	Battery	1.5	15.5 × 23 × 65	55	LTE andand Satcom	Landing	Delivery and collection, video, still and multispectral imagery, other

smart city delivery, and 5G-AI-assisted delivery. We provide detailed explanations on each of the major types of drone-based delivery that can benefit the current smart healthcare domain with a futuristic orientation.

4.1 Last mile delivery

Last mile delivery is a prominent application of medical drone delivery where drones are expected to deliver medical items to remote locations. A number of key performance indicators (KPIs) can be listed to evaluate the significance of the last mile medical delivery by a drone. For example, such KPIs can be classified into three major types: time of delivery, cost of delivery, and efficiency of the drone. Important KPIs can be considered as follows: number of stops on the fly, on-time delivery, cost per delivery, miles per watt of power, predicted versus actual distance traveled by drones, shipment accuracy, drone in motion on air, and drone in static condition on air. Route planning is an important component for KPIs of last mile delivery.

Last mile delivery use cases must be formulated against the order fulfillment versus delivery approach notions. Thus, IoT can play a vital role to support the monitoring of drones, controlling, optimization, automation, and learning of the context. Blockchain technology herein can improve the data transparency, minimal risk of fraud, immutable transactions, and secure data storage (Marco, 2018).

A three-phase decentralized drone management architecture was proposed by Ahmad et al. (2021). The architecture uses phase-1 for gathering medical data from IoT-based sensors. The data is then forwarded to other drone or blockchain platforms via drones for further processing. Drones must return to the origin when the delivery is complete. Smart contracts are used to manage the whole scenario in a seamless manner. Fig. 6.4 presents the three-phase drone delivery architecture with detailed interactions.

4.2 Collaborative and multiagent delivery

Multiagent systems (MAS) comprise autonomous agents, i.e., drones to exchange data within the group. In a study (Kapitonov et al., 2017), an MAS-based drone delivery system was proposed: autonomous intelligent robot agent (AIRA). This model helps to assist UAVs in a multitude of IoT-based applications. The AIRA system uses a docker cluster to serve cognitive services with the help of a package advisor, market analytics, and adapter analysis. The Ethereum and interplanetary file system (IPFS) are

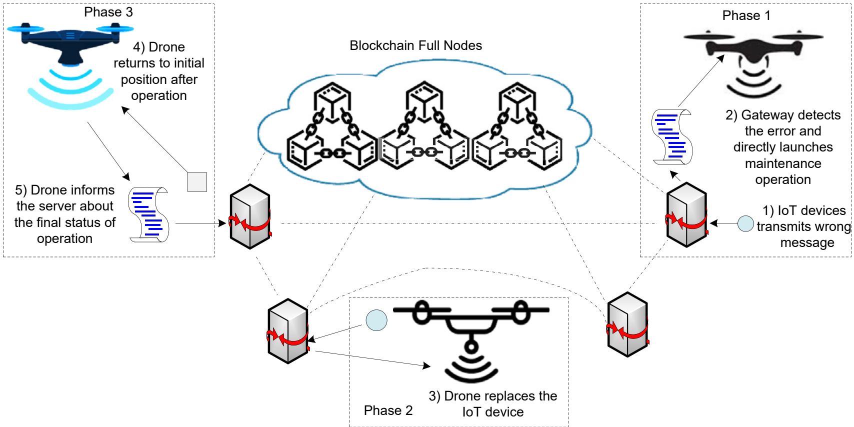

Figure 6.4 Three-phase medical drone delivery architecture.

combined to alleviate the AIRA kernels to provide seamless drone management for medical delivery purposes. AIRA uses MAS for leveraging communication with the blockchain platform via a peer-to-peer network. The system empowers the drones with efficient owner's rule and localized identity management.

Multidrone collaboration framework has become a popular approach encountered in medical delivery applications. We can use the end-to-end delivery scheme where blockchain platforms can act as the core of decentralized data storage. The system works on request and response workflow where location, trajectory, timing, and other necessary information is stored in the blockchain network. Drones can alleviate the utility of blockchains for performing various types of medical delivery scenarios. Drones that follow multirobot design need to follow the localized intelligent edge computing that allows them to decide instantly based on the gathered information about a context.

RobotChain (Alsamhi et al., 2021a,b) is such a multiagent drone architecture that uses private, public, and consortium blockchains for medical delivery. Fig. 6.5 presents the RobotChain architecture for medical drone delivery.

4.3 Epidemic support delivery

The world has witnessed the wrath of COVID-19 as a devastating epidemic. Blockchain, IoT, and drones are being used to counter COVID-19 in different parts of the world. For example, contactless delivery of medicines to COVID-19-affected patients can be done by drones. We also

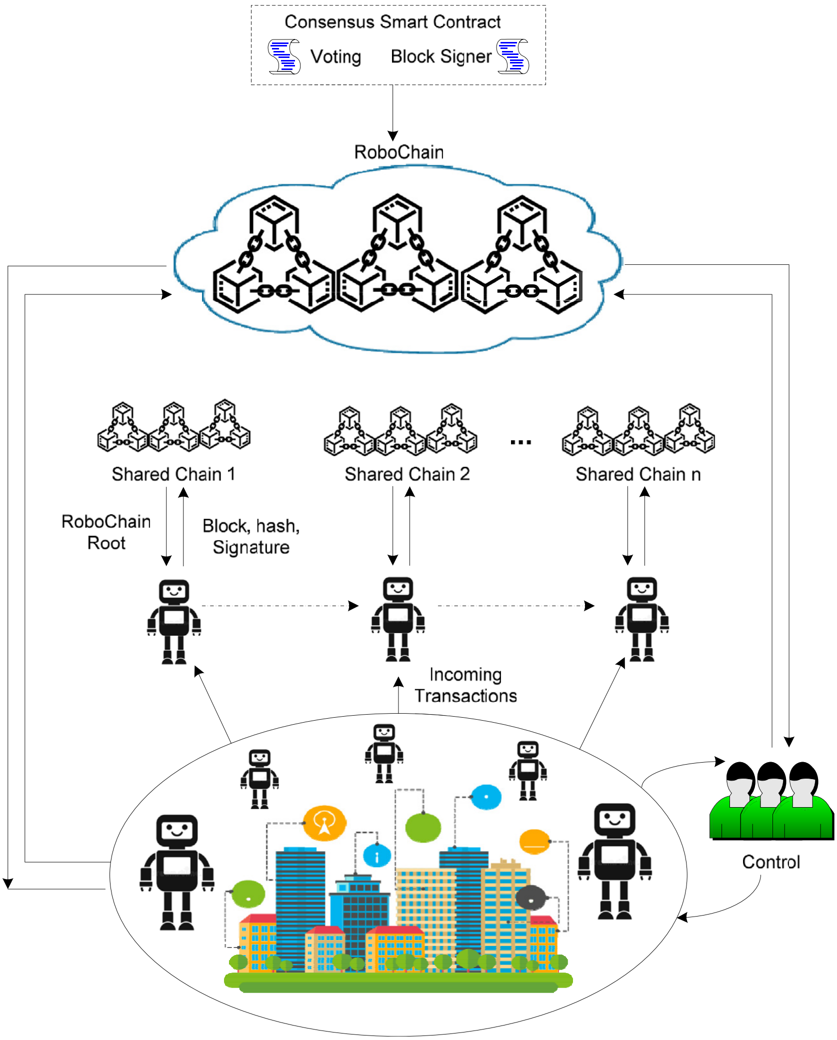

Figure 6.5 RobotChain architecture for medical drone delivery.

find that drones have been used in other applications such as food distribution during COVID-19 lockdown, delivery of vaccines by using supply chain management, COVID-19 patient information sharing, and awareness campaigning for COVID-19 (Kalla et al., 2020). Various types of consensus algorithms are used in such aspects. For instance, proof of work and proof of stake are used in contact tracing, patient information sharing, food distribution, and automated contactless surveillance. We notice that Ethereum and Hyperledger Fabric are the most used blockchain platforms in such scenarios. In Gupta et al. (2021), a UAV-based drone-centric medical delivery architecture was proposed to combat COVID-19. This architecture uses a four-layered approach considering COVID-19, virtualization, network control, and application layers. Drones collect and monitor the COVID-19 situation and forward the information to the virtual infrastructure manager. A virtual drone swarm network is envisaged herein to manage the delivery of the drones. The network control plane performs network monitoring, access control, dynamic routing, and switching between various software-defined network controllers. Smart contracts are deployed in each of the four layers, which are connected with the Ethereum blockchain. Fig. 6.6 presents software-enabled multi-swarming UAV architecture for medical delivery. In Kumar et al. (2021), a drone-based control, monitoring, and analysis architecture was proposed. The architecture uses a COVID-19-centric CC room for management of the overall drone delivery process. Drones are equipped with IoT-based data collection sensors. Drone signaling systems communicate with the centralized control room for necessary command procurement. Drones send the information to the edge, fog, or cloud platforms per their vicinity and availability of services. Thermal imagery sensors can sense whether a patient is wearing a mask and whether the patient's body temperature is within the normal range. In Firouzi et al. (2021), a number of issues are discussed in the

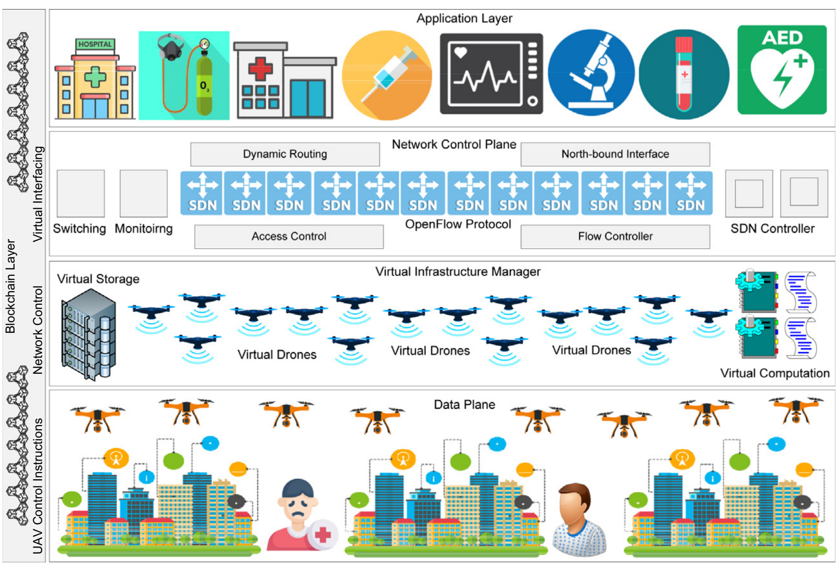

Figure 6.6 Software-enabled multi-swarming UAV medical delivery architecture.

drone-assisted COVID-19 combat scenario. Also, a pandemic-tracking smartphone app is used to update COVID-19 safety status about a patient and check the mask inventory.

4.4 Forensic delivery

Drones can be useful for both forensic and paramedical delivery operations. In Yu et al. (2019), a self-adaptive architecture called LiveBox was proposed to deal with forensic aware drone operations. In this software design, drones are equipped with a LiveBox that can be controlled from a central station. The LiveBox ground station is responsible for detecting the drones and updating the logs based on the collection information. Third-party stakeholders can access the LiveBox cloud services to check the status of the drones and perform queries. It can verify the consensus algorithm using hashing keys and make context-aware decisions. A smart contract performs recovery of flight data based on tamper-proof evidence. The LiveBox uses four-stage feedback that includes monitoring, analysis, planning, and execution. The proposed software can address two key research issues: stringent forensic application deployment ability of the drones and live forensic analysis while following regulatory corridors (no-fly zone, restricted zone, private zone, etc.). It uses a self-adaptive reporting algorithm to measure the accurate geolocations when it falls under the drone fly zone. Fig. 6.7 presents the LiveBox architecture.

4.5 Smart city delivery

Smart city, aware, drone-based medical delivery is a promising application where various types of healthcare services can be provided. For example, blockchain can be used to create a health bank that can be accessed with the help of DApps. We can also share medical data by using a medical record keeping blockchain. Blockchain can be used for the medical data storage,

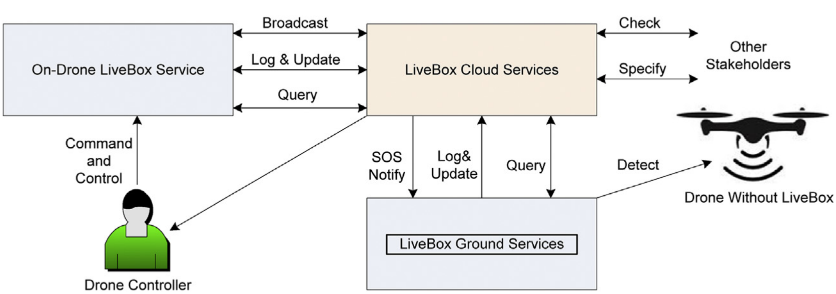

Figure 6.7 LiveBox forensic medical drone delivery architecture.

which can be useful for medical delivery use cases. Drones can be linked with the cloud-based blockchain platforms to provide such medical services in a smart city framework (Aggarwal et al., 2019). Medical professionals can access the blockchain storage with consent from the patient. Drones can be aligned with the IoT-based smart health sensors that can be used for gathering and monitoring of real-time health status.

A typical example of a smart city drone delivery use case can be described as follows. IoT-based smart sensors collect health data from the patients, which is then transmitted to the authorized drones for further forwarding to the mobile edge computing (MEC) platforms. All of the information is processed at the MECs and stored in the blockchain infrastructure. In Islam et al. (2019), a smart city, aware, blockchain architecture is proposed: BHMUS. BHMUS uses MEC to process citizen-centric health data by using drones. The drones gather data from citizens that is processed at the MEC-enabled base stations. Existing cellular networks are used to convey such data transmission between distributed MEC-enabled base stations. Each of the base stations is equipped with a blockchain that takes control of localized data storage. The whole architecture is served by standard enterprise cloud platforms. Fig. 6.8 presents a smart city—enabled drone delivery architecture.

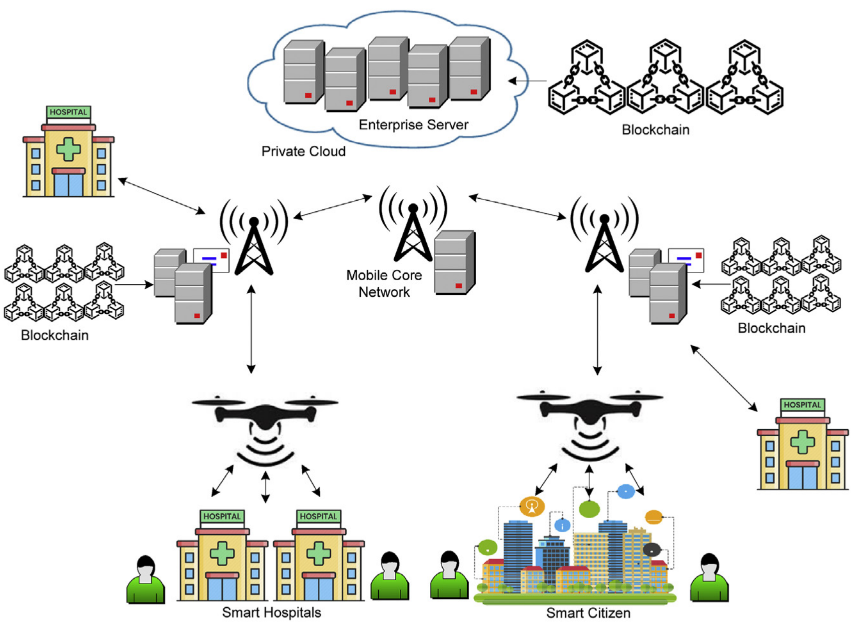

Figure 6.8 Smart city—enabled medical drone delivery and monitoring.

Smart city aware drone delivery must be safeguarded from possible threats and attacks that include cryptographic vulnerability, propagation delay, distributed denial of service attacks, infrastructure security, data integrity, data privacy, and smartphone-aware attacks (Singh et al., 2020). Smart city—based intelligent infrastructure needs to be revised in terms of the following key aspects, transmit management of drones, patient information, crash prevention, arterial management, commercial drone operation, freight management, and incident management.

4.6 5G- and AI-assisted delivery

Blockchain can be integrated with 5G and artificial intelligence (AI) to empower drone-based medical delivery. A study by Gupta et al. (2020) proposed a three-layer architecture to allow various delivery applications by using drones. A UAV layer is used to support the delivery-aware applications, e.g., healthcare, smart grid, smart city, and military surveillance. It is supported by an edge-AI layer that integrates the UAV layer to the blockchain layer. The edge-AI layer performs the decision-making based on the received information from the drones. The communication technology that is used in this architecture relies on the 5G infrastructure to minimize the latency of data transmission. Further, cloud platforms are used to analyze the filtered information that is not served by the edge-AI systems. The blockchain layer provides the security of the information collected and processed by the drones and the edge-AI layers. Blockchain allows the utilization of a UAV smart contract to communicate with the Ethereum blockchain platform. Fig. 6.9 presents the three-layer blockchain-assisted medical drone delivery architecture.

A study by Bera et al. (2020) proposed the BSD2C-IoD architecture to enable delivery operations by using drones. In this architecture, the control room sends a registration request to the registration authority (RA) and receives the acknowledgment from the RA. The prospective drone is registered with the service, which then starts delivery operations with constant communication with the ground stations. Ground stations send the block creation information to the peer-to-peer ground station network for efficient management of blocks in the remote blockchain center. This architecture uses an internet of drone (IoD) scheme for secure data delivery to remote locations.

We find key benefits of using 5G and AI in the blockchain-drone delivery ecosystem as follows: decentralized automation, scalability, identity management, autonomy, security, reliability, and secure code deployment (Mistry et al., 2020). Use of 5G and AI can improve medical delivery

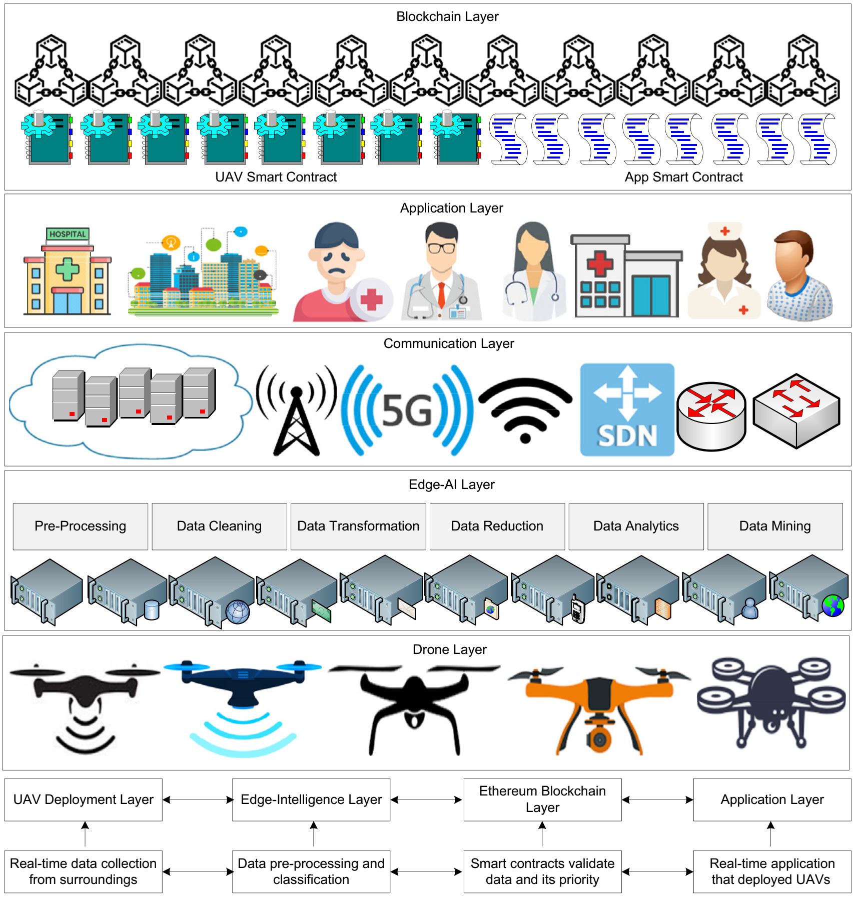

Figure 6.9 Edge-AI-assisted drone delivery architecture for medical items.

applications with the following features: dynamic deployment ability, drone swarm network, and design of aerial base stations. However, optimization of drone weight, battery capacity, and computing ability remain key issues (Han et al., 2021) (Table 6.2).

5. Key issues and future direction

5.1 Health record storage cost

Medical drone delivery can face huge costs for medical data due to storage in the blockchain platforms. At the present, very few blockchains exist that provide minimal cost per transactions (Rabah, 2018). In such a context, expenditure due to health data transactions can impact new startups and

Table 6.2 Different types of medical delivery of drones.

Application	Medical use	Medicinal items	Drone site	Advantages	Limitations
Palliative care	Delivery of controlled drugs and medicines	Injectable medicines	Remote site, outdoor	Control	Security, safety
Diagnosis	Specialized diagnostic test kits delivery	Blood, tissue, pathology samples	Indoor, clinical use	Centralized supply	Range, safety
Diagnostic	Nuclear imaging	Short half-life radionuclide	Indoor, clinical use	Rapid transfer from remote location	Safety, security
Emergency care	Overdose of medicine	Prothrombin	Outdoor, remote site	Rapid transfer to remote site	Delivery speed, cruise speed
Emergency care	Cardiac arrest	Adrenaline, lidocaine, atropine	Outdoor, remote site	Rapid transfer to remote site	Delivery speed, cruise speed
Emergency care	Asthma attack	Salbutamol, nebulizer	Outdoor, remote site	Rapid transfer to remote site	Delivery speed, cruise speed
Emergency care	Snake bite	Antivenom, antidotes	Outdoor, remote site	Rapid transfer to remote site	Delivery speed, cruise speed, range, stability
Emergency care	Burn	Live maggots	Outdoor, remote site	Centralized supply to remote site	Range
Emergency care	Clotting agent	Live leeches	Outdoor, remote site	Centralized supply to remote site	Range

Continued

Table 6.2 Different types of medical delivery of drones.—cont'd

Application	Medical use	Medicinal items	Drone site	Advantages	Limitations
Emergency care	Allergic reaction	Epi pen	Outdoor, remote site	Rapid transfer to remote site	Range, medicine stability
Emergency care	Suspected sepsis	IV antidote	Outdoor, remote site	Rapid transfer to remote site	Range, medicine stability
Emergency care	Chemotherapy	Short half-life radionuclide	Indoor, clinical use	Rapid transfer from remote site	Security, safety
Emergency care	Disease outbreak	Vaccine	Outdoor, remote location	Remote delivery	Range, stability, medicine stability
Emergency care	Organ delivery	Heart, kidney, eye, liver	Outdoor, remote location, hospitals	Rapid transfer to remote site	Range, organ stability, speed of delivery, guarantee of delivery
Emergency care	Blood delivery	Blood packets	Outdoor, remote location, hospitals	Rapid transfer to remote site	Range, blood stability, speed of delivery, guarantee of delivery
Regular use	Patient guidance	Medical support with voice	Outdoor, hospital out area	Centralized control	Stability

technology deployments. Novel techniques should be sought to use public blockchains, but proper security and privacy issues need to be addressed beforehand.

5.2 Energy management

Energy consumption by the drones during flight and charging could be a major problem in the coming days. Normally, drones get charged by standard alternating current–centric electrical modules. This can indirectly increase fossil fuel consumption. New methods must be developed that can use renewable energy sources to charge the UAVs. However, cost and circuit complexity should be optimized simultaneously.

5.3 Ecologic condition

Use of drones for medical delivery can populate open air space that can hinder the normal working of natural ecology. Thus, ecologic conditions must be balanced before deployment of drones for medical delivery. We should focus on the dismantling of drones when they expire. Proper renewal strategy should be provided before using drones for healthcare applications.

5.4 Legal disputes

Drones may enter such locations or geographic zones that are either restricted or avoided, e.g., military bases or industrial buildings. In that case, the drone operators may face legal issues (Jain et al., 2019). Drone suppliers and operators must follow strict geographic fencing algorithms to safeguard the drones from possible legal disputes. Further, drones should be programmed to handle medicinal products and deliver the shipments to appropriate destinations. Notwithstanding these points, drone suppliers may face legal disputes.

5.5 Latency and throughput

Drone-based medical delivery needs to have latency-aware systems. For example, emergency organ delivery applications should avoid any type of latency (network or speed) to avoid health hazards to the patient. Throughput of drone-based delivery depends on the amount of delivery successfully completed against a given time window. Thus, drone operators should follow a priority basis medical delivery scheme to minimize the issues related to latency.

5.6 Resource utilization

IoT works on a resource-constrained framework. Thus, IoT- enabled drone networks must be aware of resource-constrained design. We should find solutions that can mitigate and optimize the existing resource utilization for medical delivery. Drones should not over depend on IoT-based resources to make them compulsorily work in an overclocked manner. Further solutions should be sought to look into the underutilization of the available resources.

5.7 Vulnerability

Drones are expected to fly in the open air and remote fields. In such a context, drones can face vulnerable conditions by manual attack or military fire. Drones may also be considered in the scenario where climatic conditions are against the proper workflow of their in-built mandates. Thus, vulnerable prediction analytics and recovery of drones need to be incorporated into the network design (Alladi et al., 2020). Blockchains can help to account for tamper-proof data storage to eradicate the vulnerable behavior of humans or nature to safeguard private health information.

5.8 Standard architecture

There is no standard architecture that exists to date that can guide the proper deployment of drones under the aegis of blockchain and IoT. In such a context, initiatives should be taken by international organizations to come up with a standard architecture following a rigid protocol-aware formulation.

5.9 Incentive for participation

Ordinarily, blockchains provide incentives to miners in terms of cryptocurrency, i.e., economic benefit. Drone-aware medical delivery should be empowered with incentivization schemes for all stakeholders (Al-Marridi et al., 2021; Xu et al., 2021; Manu et al., 2021; Johari et al., 2021). It would boost their involvement to contribute for safety, privacy, and vulnerability-free passage through the open air. For example, a miner who gives prior information about a possible obstacle or vulnerable attack can be given a token of cryptocurrency. Thus, a holistic ecosystem can be devised where stakeholders from all categories can join the blockchain and improve the delivery service.

5.10 Regulatory constraints

To date, there are no fixed regulations that are applicable in all the countries of the world. This hinders the process of deployment and utilization of various bands of frequency spectrum in different locations (Sujihelen, 2021; Reinert and Corser, 2021; Sharma et al., 2021; Chouhan et al., 2021). Further, there is no fixed rule for hovering drones in the open air. Some countries allow drones to fly at 5000 m above sea level. But sometimes, it may not be possible due to some technical or administrative reasons. Further, consideration should be paved to include insurance companies to compensate for accidents or destruction of drones while on duty for delivery. We expect that soon policy makers from all countries will design regulations that can be modified based on a consensus algorithm mechanism.

5.11 Technology adoption

Adoption of technology for medical use cases is not an easy task. It is often observed that people from different educational backgrounds and economic stature take such matters in a completely unexpected manner. Utilization of drones for medical delivery can face social inertia that can hinder the acceptance of such technology (Ch et al., 2020). We expect that proper campaigning and awareness about public safety can ensure the minimization of resistance to technology adoption in society.

5.12 Future direction

Vulnerability prevention should be the first step toward futuristic drone-centric medical delivery applications. We should focus on various levels of vulnerabilities, e.g., communication channel level, transceiver level, and control center level. Use of fuzzy logic and advanced DL can benefit vulnerability prediction, detection, and recovery simultaneously (Eichleay et al., 2019). We need to further think of using layered blockchain architecture where the workflow should start from the data layer, i.e., gathering data from IoT-based devices. The information should then be traversed to the next layers, such as the network layer. It can work as the key to all communication activities and cover all types of heterogeneity protocols to allow IoT devices to cooperate. Higher layers of such blockchain architecture may consist of consensus algorithm layer, incentive layer, smart contract layer, and application layer. Various consensus algorithms must be incorporated into the consensus algorithm layer to provide a flexible approach to serve a multitude of medical delivery applications (Thiels et al., 2015; Razdan and Sharma, 2021; Barnawi et al., 2021). An insurance mechanism and allocation of cryptocurrency token can be accommodated in the incentive layer. Smart contracts can be deployed in the contract layer, which can finally reach to the application layer, where the actual medical delivery use case would take place. Emphasis should be given to the following aspects to improve the existing drone delivery ecosystems: creation of new business models, collision-free UAV movement guarantee, uniform load (medical data) sharing (Ling and Draghic, 2019), data entity authentication, and lightweight consensus algorithm design (Rahman et al., 2021; Irshad et al., 2021). Upcoming drone delivery startups may need to fulfill cooperation and faster synchronization schemes between drones and remote CC centers. Decentralized decision-making would be another important avenue that can enhance the acceptance and availability of drone-based health services. Utilization of multidrone (swarms) techniques can alleviate the channel utilization with enriched security and transparency (Du et al., 2021; Saxena et al., 2021; Guggenberger et al., 2021). UAV-based cloud or fog platforms can be considered as another future direction to make medical delivery accessible to every citizen. Internet agnostic network design techniques could expand the reachability of the drones for efficient deployment at the extreme last mile scenario. Dew computing paradigm could be considered the perfect match for devising the dew-IoT drone cluster implementation that merely depends on the internet backhaul for exchange of data (Ray, 2017). Virtual circuit-based risk assessment models may be investigated for eradication of unprecedented failure of drones due to technical issues.

6. Conclusion

In this article, we present an in-depth review of drones for possible deployment in the medical delivery use cases. We envisage that drones would be accepted to encounter the gradually increasing requirement of medical delivery at lower cost and higher security than what exists today. We expect that proper incorporation of lightweight blockchain-based frameworks may harness transparency as well as privacy, alleviating medical delivery applications. Existing drone suppliers are doing fantastic jobs to lift up drone-assisted medical care, though because of lack of latency-mitigating technologies and proper planning, they seem to be resisted by general social inertia. It is however understood that upon certain changes in the socio-technologic paradigm, drones are certainly going to revolutionize the delivery domain to make healthcare smarter.

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